



EXPERNETIC (PVT) LTD

Talent Assessment Program · Technical Assignment

Data Engineer Intern

Final Report - Bangkok Airbnb Market Intelligence

Real-World Data Challenge - Inside Airbnb Dataset

Field	Detail
Candidate	Ishini Ellewela
Role Applied For	Data Engineer Intern
Dataset	Inside Airbnb - Bangkok, Thailand (listings, calendar, reviews, neighbourhoods)
Report Type	Comprehensive analytical & engineering report (Section 12 deliverable)
Source Artifacts	4 Jupyter notebooks (Sections 2, 3, 4, 6) + 8 exported chart figures
Report Format	PDF · Compiled from candidate notebooks

Table of Contents

1. Executive Summary.....	5
Headline Findings	5
Scope & Prioritization.....	5
2. Objectives & Scope	7
2.1 Assignment Objective.....	7
2.2 City Selection	7
2.3 Prioritization Rationale.....	7
3. Dataset Overview.....	8
3.1 Source.....	8
3.2 Files Used.....	8
3.3 Entity Relationships.....	8
3.4 Business Domain Context.....	8
3.5 Schema Profiling Highlights (listings.csv.gz, raw).....	9
3.6 Dataset Limitations Identified	9
3.7 Assumptions Documented by the Candidate	10
4. Methodology.....	11
4.1 Tooling.....	11
4.2 Reasoning Behind Key Method Choices.....	12
5. Engineering Approach.....	13
5.1 Pipeline Overview.....	13
5.2 Ingestion & Profiling Results	13
5.3 Cleaning & Standardization Steps	13
5.4 Enrichment & Joining	14
5.5 Data Quality & Validation Checks	14
5.6 SQL Analytical Layer	15
5.7 Engineering Decision Log	15
6. Exploratory Data Analysis Findings.....	17
6.1 Price Distribution & Room Type.....	17
6.2 Geographic & Neighbourhood Analysis	18
6.3 Review Scores, Room-Type Mix & Host Concentration	19
6.4 Availability & Occupancy	20

6.5 Temporal Trends — Review Volume 2015–2026.....	21
6.6 Sections Not Completed: Geographic Mapping & Formal Clustering	21
7. Statistical Findings.....	22
7.1 What the Regression Model Implicitly Suggests About H1–H5	22
8. Data Science Experiments — Price Prediction.....	23
8.1 Problem Framing	23
8.2 Feature Engineering	23
8.3 Train/Test Split	24
8.4 Model Comparison	24
8.5 Feature Importance.....	25
8.6 Residual Analysis	26
8.7 Business Applications & Limitations (as documented by the candidate)	27
8.8 Methodological Gaps vs. the Assignment Brief	27
9. AI/ML Experiments (NLP, LLM, Agentic)	28
10. Visualizations — List of Figures.....	29
11. Business Recommendations	30
11.1 For Revenue Strategists / Pricing Teams.....	30
11.2 For Operations / Marketplace Health	30
11.3 For Product / Host Growth.....	30
11.4 For Investors / Market Entrants	30
12. Cross-City Comparisons	31
13. Limitations & Caveats	32
13.1 Data Limitations	32
13.2 A Specific Data-Quality Issue Found During Review	32
13.3 A Narrative/Result Consistency Issue (Corrected in This Report).....	32
13.4 A Minor Numerical Inconsistency (Corrected in This Report)	33
13.5 Engineering & Modelling Scope Limitations	33
13.6 Statistical / Practical Significance	33
14. Future Improvements	34
14.1 Quick Wins (hours, using data already ingested).....	34
14.2 Medium Effort (a few days)	34
14.3 Larger Investments (a week or more)	35
15. Reflection	36

15.1 Prioritization	36
15.2 Trade-offs Made	36
15.3 Key Lessons Learned	36
Appendix A: AI Usage Disclosure	37

1. Executive Summary

This report presents the findings of a market intelligence study of the **Bangkok, Thailand** short-term rental market, built on the public Inside Airbnb dataset. The analysis spans four completed workstreams: dataset familiarization, data engineering, exploratory data analysis, and machine learning price prediction covering **31,069 raw listings, 693,647 guest reviews, and over 11.3 million calendar-day records.**

Headline Findings

- **Market size & structure:** 28,204 listings remained after price cleaning (91% of raw data). Entire homes/apartments dominate supply (66.2%), followed by private rooms (31.1%). The market is highly concentrated: three neighbourhoods - Vadhana, Huai Khwang, and Khlong Toei — account for roughly 36% of all listings.
- **Pricing:** Median nightly price is ฿1,549 (mean ฿2,031), a right-skewed distribution typical of accommodation markets. Parthum Wan (central Bangkok, ฿2,547 median) is the premium hub; Hotel rooms command the highest median price by room type (฿1,940).
- **Supply utilization:** Median booked-day estimate is only 6 days per year against 337 median available days — an estimated occupancy rate of just 1.6%. This points to a large share of inactive or "ghost" listings inflating headline supply figures.
- **Host economics:** Superhosts (35.6% of listings) charge a measurable premium and earn materially higher review volumes and scores than regular hosts, indicating the Superhost badge carries real monetary value.
- **Demand recovery:** Review volume used as a demand proxy collapsed during COVID-19 (2020–2021) and has since recovered to roughly 3x pre-pandemic (2019) levels by 2025, signalling strong post-pandemic tourism growth in Bangkok.
- **Predictive modelling:** A Random Forest regressor was the best of three model families tested, explaining 66.4% of price variance ($R^2 = 0.664$) with a mean absolute error of ฿551/night. Bedrooms, neighbourhood price level, and host portfolio size were the dominant predictors.

Scope & Prioritization

Per the assignment's stated design philosophy that depth on fewer sections outperforms shallow coverage of all sections the candidate concentrated effort on Sections 2 (Dataset Familiarization), 3 (Data Engineering, core layer), 4 (Exploratory Data Analysis), and 6.1 (Price Prediction). Sections 5 (formal hypothesis testing), 6.2–6.4 (forecasting, clustering, bias testing), 7 (NLP/LLM/agentic experimentation), and 8 (Open Innovation) were not attempted in the current submission; these

are addressed transparently in Sections 13–15 of this report, with a rationale and a concrete plan for completing them.

Business Recommendations at a Glance

- Deploy the Random Forest price model as an advisory pricing tool for new hosts, with human review given the ฿551 average error band.
- Investigate the very low estimated occupancy rate (1.6%) before using "listing count" as a proxy for market health a large share of supply may be dormant.
- Use neighbourhood-level median pricing (Parthum Wan, Vadhana, Samphanthawong) to guide new-host pricing guidance and investor site selection.
- Continue to incentivize Superhost status given its association with materially higher prices, review volume, and guest satisfaction.

2. Objectives & Scope

2.1 Assignment Objective

The brief (Expernetic Data Engineer Intern Assignment, Real-World Data Challenge) asked the candidate to act as a data engineer and analyst supporting a hypothetical Airbnb market-intelligence consultancy, transforming the raw Inside Airbnb dataset into engineering artifacts, analytical insights, and business recommendations for product, revenue, and operations stakeholders.

2.2 City Selection

A single-city scope **Bangkok, Thailand** was selected. Per Section 9 of the brief's City Selection Guide, a single-city approach is intended to demonstrate depth, code quality, and storytelling, mastering one market before generalizing. Bangkok was a strong choice: it is one of the largest and most data-rich cities in the Inside Airbnb catalogue (31k+ listings, near-complete calendar coverage, ~694k reviews) and offers a clean before/after COVID-19 demand shock to analyze.

2.3 Prioritization Rationale

Given the assignment's explicit statement that it is scoped beyond what can be completed in one week, the candidate prioritized as follows:

Priority	Section	Status	Rationale
1 (Mandatory)	02 — Dataset Familiarization	Complete	Foundation for all downstream work; required before any modelling.
2 (Recommended)	03 — Data Engineering (3.1–3.3)	Complete	Ingestion, cleaning, enrichment and a queryable SQL layer needed for EDA/ML.
3 (Recommended)	04 — Exploratory Data Analysis	Complete	Directly answers stakeholder questions on pricing, geography, hosts and demand.
4 (Optional)	06.1 — Price Prediction	Complete	Highest-value optional deliverable; demonstrates full ML lifecycle end-to-end.
Deprioritized	03.4–3.6, 05, 06.2–06.4, 07, 08	Not attempted	Traded breadth for depth on the four sections above;

Priority	Section	Status	Rationale
			documented honestly in Sections 13–15.

3. Dataset Overview

3.1 Source

All data originates from Inside Airbnb (insideairbnb.com), an independent, publicly available project that scrapes and republishes Airbnb listing, calendar, and review data for cities worldwide. No supplementary or external data sources were used, consistent with the assignment's constraints.

3.2 Files Used

File	Rows	Columns	Description
listings.csv.gz	31,069	90	Core listing attributes: price, room type, location, host info, amenities.
reviews.csv.gz	693,647	6	Guest review text with reviewer ID, date, and listing reference.
calendar.csv.gz	11,340,155	5	Daily price/availability status per listing (~365 days each).
neighbourhoods.csv	50	2	Bangkok's 50 official districts (khet) and (empty) neighbourhood groupings.

3.3 Entity Relationships

- listings.id → reviews.listing_id (one listing : many reviews)
- listings.id → calendar.listing_id (one listing : ~365 calendar-day rows)
- listings.neighbourhood_cleansed → neighbourhoods.neighbourhood (listing : district)

3.4 Business Domain Context

- **Listing:** a bookable unit (private room, shared room, hotel room, or entire home/apartment) with fixed attributes (location, capacity, amenities) and a variable price.
- **Host:** the individual or commercial operator who owns/manages one or more listings; hosts range from single-unit casual hosts to portfolio operators managing 20+ properties.

- **Review:** a guest-submitted, timestamped record tied to a completed stay; review volume over time is used in this report as a proxy for realized demand/bookings, since actual booking/transaction data is not published by Airbnb.
- **Calendar day:** a single date's availability status (booked/blocked vs. open) and applicable minimum/maximum night rules for one listing.

3.5 Schema Profiling Highlights (listings.csv.gz, raw)

The candidate's Section 2 notebook profiled all 90 columns for data type, non-null count, missing percentage, and a sample value. Selected findings:

Column	Missing %	Note
neighborhood_overview	100%	Field entirely unpopulated in this scrape.
host_since	100%	Entirely unpopulated — see Section 13 for the modelling implication.
host_response_time / rate, host_acceptance_rate	100%	Airbnb no longer exposes these via scrape.
license, instant_bookable, calendar_updated	100%	Not populated for Bangkok in this scrape.
review_scores_rating (and sub-scores)	32.2%	Missing for listings with zero reviews (expected, not an error).
host_location	25.8%	Hosts choosing not to disclose location.
price	6.7% (raw)	Combined with out-of-range values, 2,865 rows (9.2%) were ultimately excluded.

3.6 Dataset Limitations Identified

- 14 columns are 100% empty for this city/scrape these were excluded from modelling and enrichment rather than imputed, since imputing an entirely-missing column manufactures information that does not exist.
- Price is scraped from the listing page at time of scrape and may not reflect the guest's final paid price after discounts, cleaning fees, or currency-conversion artifacts.
- Review volume is a lagging and incomplete proxy for demand not every stay generates a review, and review propensity may vary by host type or guest nationality.

- The dataset is a single point-in-time scrape (last_scraped $\approx$ 2026-07-02); it reflects a snapshot, not a longitudinal panel, other than the historical calendar and review-date fields.

3.7 Assumptions Documented by the Candidate

- Prices outside the range (£0, £10,000) were treated as data errors and excluded (2,865 rows / 9.2% of raw listings) see Decision Log, Section 5.4.
- Missing review_scores_rating was treated as "no reviews yet" rather than "poor quality," and was either left null for descriptive statistics or median-imputed only for the modelling feature matrix.
- host_is_superhost and similar boolean text flags ('t'/'f') were mapped to native booleans/1-0 integers for downstream use.

4. Methodology

The overall analytical approach followed a standard applied-analytics lifecycle, executed across four Jupyter notebooks, one per assignment section:

Notebook	Assignment Section	Purpose
01_section2_dataset_familiarization.ipynb	Section 2	Load, profile, and document all four source files; identify relationships and limitations.
02_section3_data_engineering.ipynb	Section 3	Clean, standardize, enrich, and join data; build a queryable SQL layer; log engineering decisions.
03_section4_eda.ipynb	Section 4	Univariate, geographic, temporal, host- and review-side exploratory analysis with business interpretation.
04_section6_price_prediction.ipynb	Section 6.1	Feature engineering and comparison of three regression model families for price prediction.

4.1 Tooling

Python 3 with pandas, NumPy, matplotlib, and seaborn for data wrangling and visualization; scikit-learn for modelling (LinearRegression, RandomForestRegressor, GradientBoostingRegressor); SQLite (in-memory, via Python's sqlite3) as a lightweight analytical SQL layer. No cloud infrastructure, orchestration tool, or containerization was used in this iteration see Section 13 for the trade-offs this implies at production scale.

4.2 Reasoning Behind Key Method Choices

- **In-memory pandas over Dask/Spark:** the full working set (~500MB across all four files) fits comfortably in memory, so distributed processing would add operational overhead without a performance benefit at this scale.
- **SQLite over PostgreSQL/DuckDB:** a zero-install, single-file engine was sufficient to demonstrate SQL competency (Query 1–5) without requiring the grader to stand up external infrastructure.
- **Train/test split over pure descriptive modelling:** an 80/20 split with a fixed `random_state=42` was used for the price-prediction task to produce a reproducible, held-out performance estimate.
- **Three model families compared:** a linear baseline (Linear Regression), a bagging ensemble (Random Forest), and a boosting ensemble (Gradient Boosting) spanning the bias/variance spectrum typically seen in tabular price-prediction tasks.

5. Engineering Approach

5.1 Pipeline Overview

The engineering notebook (Section 3) implements a linear, notebook-driven pipeline: Ingest → Profile → Clean → Standardize → Enrich/Join → Validate → Load to SQL → Query → Persist. This satisfies the assignment's "Data Ingestion & Profiling," "Data Cleaning & Standardization," and "Data Enrichment & Joining" sub-sections (3.1–3.3) in full. The optional/advanced sub-sections - dimensional modelling (3.4), automated/incremental pipeline design (3.5), and cloud-native topics (3.6) were not attempted; see Section 13.5 for the reasoning and a proposed design.

5.2 Ingestion & Profiling Results

Dataset	Rows	Columns	Memory	Avg. Missing %	Duplicate Rows
listings	31,069	90	155.9 MB	20.8%	0
reviews	693,647	6	329.8 MB	0.0%	0
calendar	11,340,155	5	1,438.4 MB	0.0%	0
neighbourhoods	50	2	<0.1 MB	50.0%	0

No duplicate rows were found in any of the four source files. The neighbourhoods file's 50% average missingness is entirely attributable to the `neighbourhood_group` column, which is unpopulated for Bangkok.

5.3 Cleaning & Standardization Steps

- Price: stripped currency symbols and thousands separators, then cast string → float (`price_clean`).
- Dates: `host_since`, `first_review`, `last_review`, `last_scraped` parsed to native datetime.
- Booleans: `host_is_superhost`, `host_has_profile_pic`, `host_identity_verified`, `instant_bookable` mapped from Airbnb's 't'/'f' string convention to native booleans.
- Percentage fields (`host_response_rate`, `host_acceptance_rate`): verified already numeric — no transformation required (though effectively unusable given they are 100% missing for this scrape).
- Invalid price removal: rows with `price ≤ ₪0` or `price ≥ ₪10,000` were dropped **2,865 rows removed (9.2% of raw listings)**, leaving 28,204 clean listings.
- Calendar: date parsed to datetime; `available` flag mapped 't'/'f' → boolean.

5.4 Enrichment & Joining

A single enriched_listings.csv (28,204 rows × 102 columns) was produced via three successive left joins against the cleaned listings table:

- + review_summary (per-listing total_reviews, first/last review date, average comment length) - derived by grouping reviews.csv.gz by listing_id.
- + calendar_summary (per-listing total_available_days, total_booked_days, occupancy_rate) - derived by grouping calendar.csv.gz by listing_id.
- + neighbourhoods (district-level join key) on neighbourhood_cleansed = neighbourhood.

Left joins were used throughout to preserve every listing even where no reviews or calendar rows existed, at the cost of introducing nulls that downstream analysis and modelling had to handle explicitly (e.g., median-imputing review scores for the ML feature matrix).

5.5 Data Quality & Validation Checks

Check	Result
Negative prices	0
Prices > ฿10,000	0
Invalid coordinates (outside Bangkok's latitude band)	0
Duplicate listing IDs	0
Review scores > 5	0
Availability > 365 days	0

All six domain-expectation checks passed on the cleaned dataset, giving confidence that the cleaning pipeline correctly removed the problematic rows rather than merely masking them.

5.6 SQL Analytical Layer

The cleaned listings, reviews, and calendar_summary tables were loaded into an in-memory SQLite database and queried directly to answer five stakeholder-relevant questions:

- Query 1 - Average price by room type (Entire home/apt ฿2,179 avg, Hotel room ฿2,162, Private room ฿1,795, Shared room ฿599).
- Query 2 - Top 10 neighbourhoods by rated-listing count and average rating (filtered to listings with ≥1 review; Vadhana leads with 2,971 rated listings).
- Query 3 - Superhost vs. regular host performance: Superhosts average ฿2,155/night vs. ฿1,963 for regular hosts, with a materially higher average rating (4.86 vs. 4.55) and review count (38.6 vs. 14.7).
- Query 4 - Top 10 highest-rated, most-reviewed listings, dominated by Khlong Toei and Phra Khanong entire-home properties near BTS (Skytrain) stations.
- Query 5 - Commercial host segmentation by portfolio size: hosts with 20+ listings form the largest segment (10,897 listings, avg ฿1,918/night, 4.63 rating), followed by 6–20 listings (7,505, ฿2,103, 4.72), 2–5 listings (6,048, ฿2,075, 4.75), and single-listing hosts (3,754, ฿2,146, 4.71). Notably, single-listing casual hosts achieve slightly higher average prices and ratings than large commercial operators, suggesting portfolio scale alone doesn't drive premium pricing or guest satisfaction.

5.7 Engineering Decision Log

Decision	Options Considered	Choice	Trade-off Accepted
SQL engine	PostgreSQL, DuckDB, SQLite	SQLite (in-memory)	PostgreSQL needed a running server the grader would have to install; DuckDB is a strong analytical alternative but adds a dependency for marginal benefit at this data volume (~28k listing rows). SQLite ships in Python's standard library, requires zero setup, and is more than sufficient for the five demonstration queries. Trade-off: no concurrent writes and not production-grade at real

Decision	Options Considered	Choice	Trade-off Accepted
			scale — acceptable for a single-analyst, read-heavy workload.
Join strategy	Inner join, left join	Left join	Introduces NULLs for listings without reviews/calendar, but avoids silently dropping listings.
Price outlier threshold	IQR method, percentile-based, fixed threshold	Fixed ₱10,000 ceiling	Simple and interpretable, but may exclude a small number of genuine luxury listings.
Processing model	Chunked loading, Dask, in-memory pandas	In-memory pandas	Not scalable to 10x data volume without re-architecture, but simplest for a ~500MB working set.

Engineering Note

This decision log satisfies the brief's explicit instruction that interviewers will focus most of their attention here. Each entry states the options considered, the choice made, and the trade-off accepted, as required.

6. Exploratory Data Analysis Findings

All figures below are reproduced directly from the candidate's 03_section4_eda.ipynb notebook outputs. Each is accompanied by the notebook's own printed business interpretation, per the brief's Business Interpretation Requirement (Section 4, callout box).

6.1 Price Distribution & Room Type

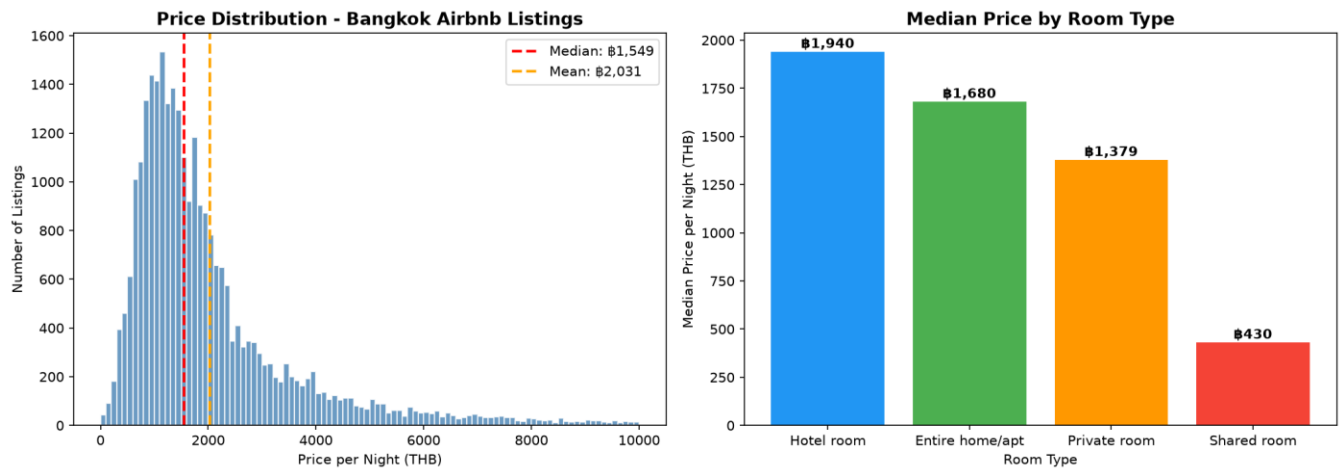


Figure 1. Price distribution across all cleaned Bangkok listings, and median price by room type.

Prices range ฿1–฿9,999 with mean ฿2,031 and median ฿1,549 (std. dev. ฿1,585) a classic right-skewed accommodation-price distribution. By room type, Hotel room listings carry the highest median (฿1,940), ahead of Entire home/apt (฿1,680), Private room (฿1,379), and Shared room (฿430).

Business interpretation (candidate's notebook): "most listings are affordable but luxury properties pull the mean up" implying that median, not mean, should anchor any market-average pricing communicated to hosts or investors, since the mean overstates what a typical guest pays.

6.2 Geographic & Neighbourhood Analysis

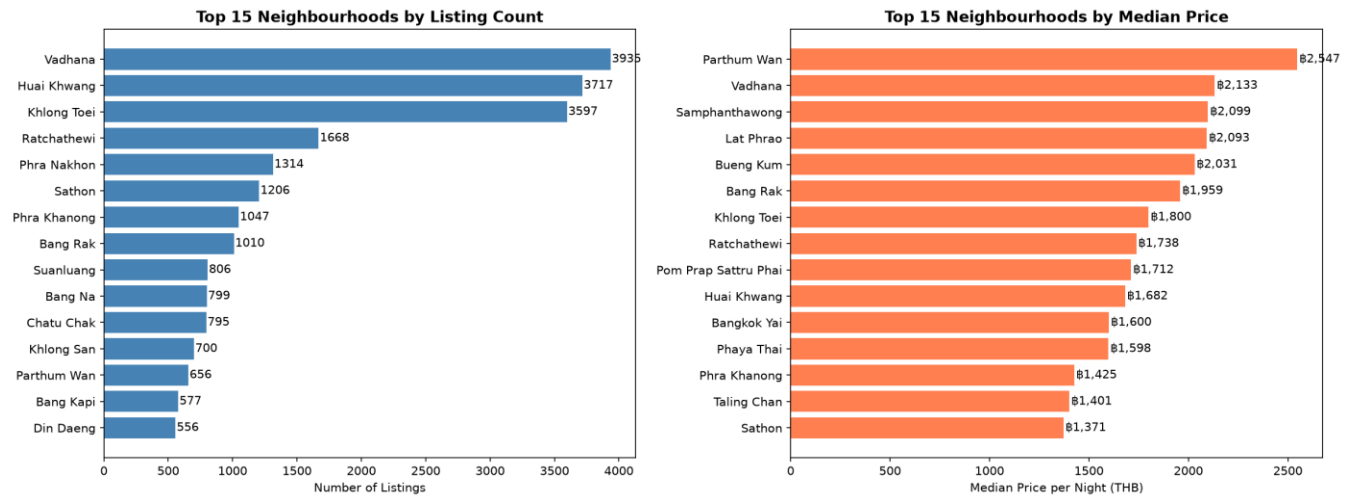


Figure 2. Top 15 neighbourhoods by listing volume (left) and by median price (right).

Vadhana is Bangkok's largest Airbnb market by volume (3,935 listings), followed by Huai Khwang (3,717) and Khlong Toei (3,597) together roughly 36% of all cleaned listings. Pricing tells a different story: Parthum Wan commands the highest median price (฿2,547), ahead of Vadhana (฿2,133) and Samphanthawong (฿2,099), while high-volume Sathon sits at the bottom of the top-15 price ranking (฿1,371).

Business interpretation: volume and price leadership do not coincide. Parthum Wan (Bangkok's central business/shopping district) behaves as the premium hub, while Vadhana's volume leadership likely reflects breadth of inventory (expat/tourist housing stock) rather than the highest achievable nightly rate. Investors optimizing for yield vs. occupancy should weigh these separately by neighbourhood.

6.3 Review Scores, Room-Type Mix & Host Concentration

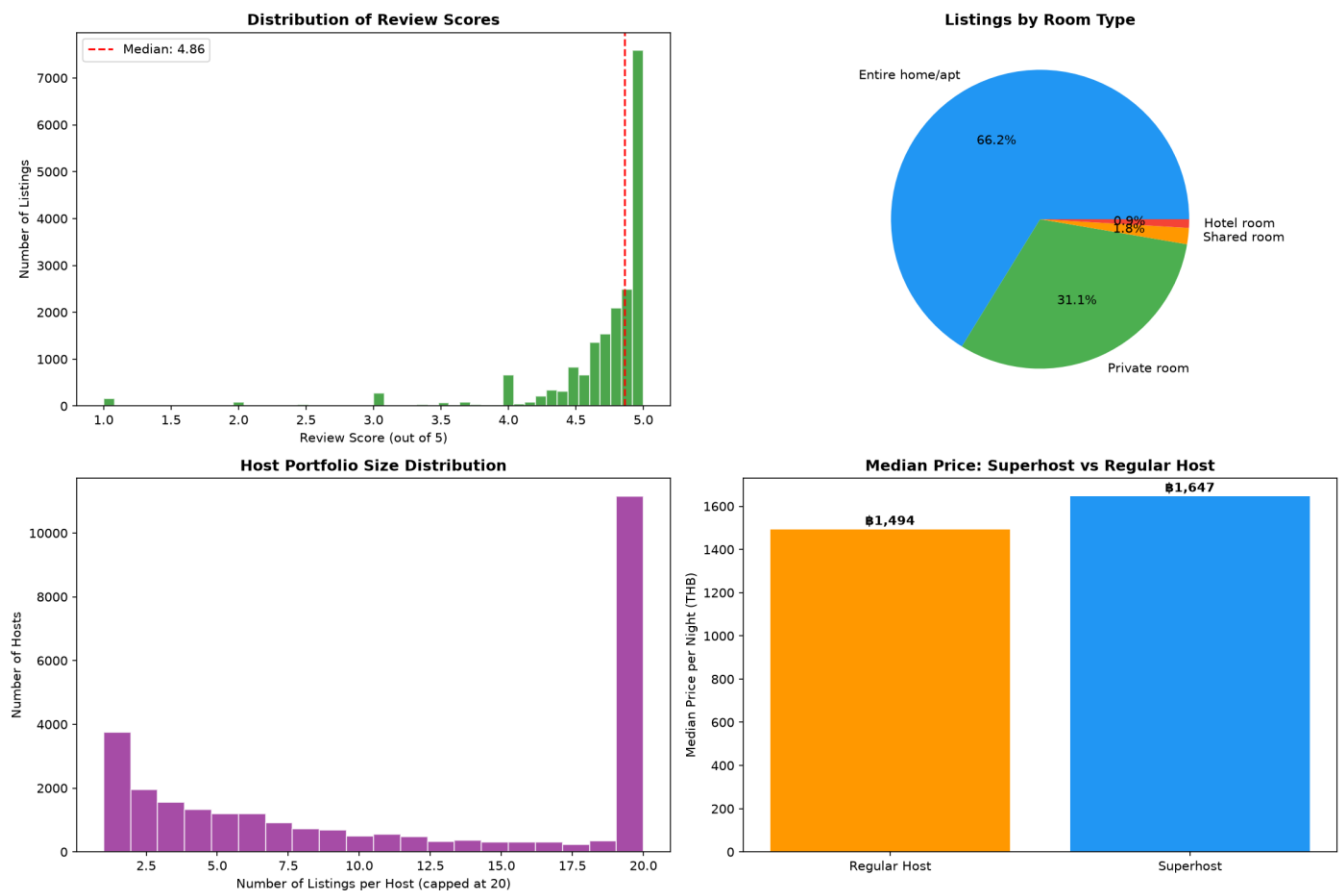


Figure 3. Review score distribution, listing mix by room type, host portfolio-size distribution, and Superhost price premium.

67.8% of cleaned listings (19,113) carry at least one review, with a median score of 4.86/5 and a mean of 4.69/5. Scores cluster tightly near the top of the scale a pattern consistent with rating inflation commonly observed on two-sided reputation marketplaces, where guests under-report middling experiences. By room type, entire home/apt listings dominate (66.2%), followed by private rooms (31.1%); hotel and shared rooms are marginal (1.8% and 0.9% respectively). The host portfolio-size histogram is heavily right-skewed and capped at 20+ listings for a large cluster of hosts, indicating a meaningful commercial/multi-listing operator segment alongside single-listing casual hosts.

Superhosts (35.6% of listings) charge a measurable premium either way it is measured: **median** nightly price ฿1,647 vs. ฿1,494 for regular hosts (a ฿153 gap, per Figure 3), and **mean** nightly price ฿2,155 vs. ฿1,963 (a ฿192 gap, per the Section 5.6 SQL query). The two figures are not in

conflict one is a median comparison, the other a mean comparison but should always be cited with the metric labelled to avoid the appearance of an inconsistency.

Business interpretation: the compressed review-score range means small absolute differences (e.g., 4.7 vs. 4.9) may represent economically meaningful quality gaps that a naive linear treatment of review_scores_rating would understate. The consistent, metric-labelled price gap for Superhosts (~฿153 median / ~฿192 mean) supports a platform recommendation to keep the Superhost incentive structure prominent.

6.4 Availability & Occupancy

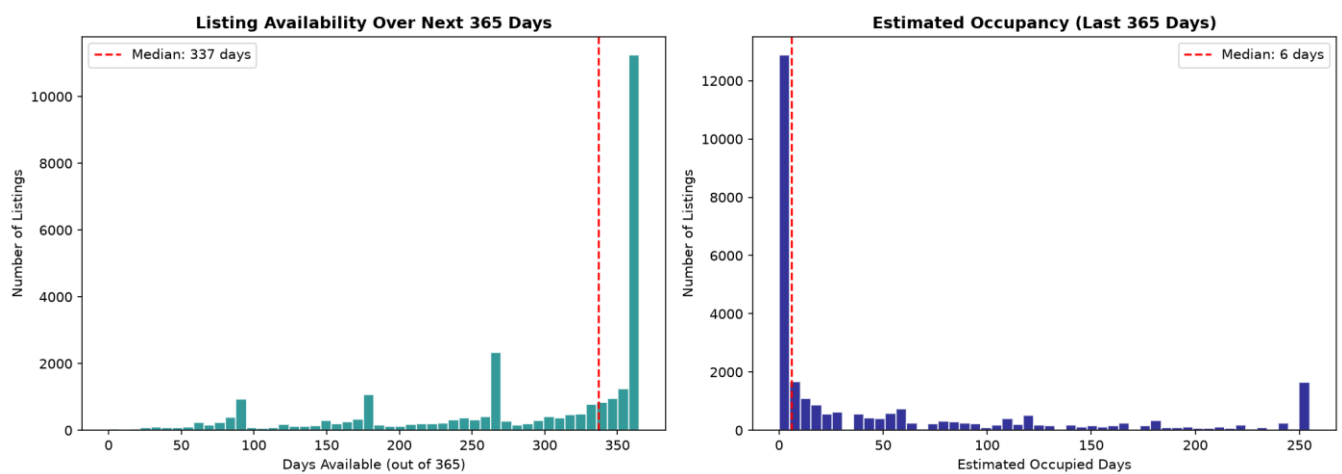


Figure 4. Days available in the next 365 days (left) vs. estimated occupied days in the last 365 days (right).

Median availability is 337 of the next 365 days most listings are open for booking almost year-round yet median estimated occupancy over the last 365 days is only 6 days, an implied occupancy rate of roughly 1.6%. 5,502 listings show full 365-day availability, and only 2 listings show zero availability.

Business interpretation: a large share of Bangkok's nominal Airbnb supply appears to be dormant or "ghost" inventory listed but rarely booked. This has two direct implications: (1) raw listing counts overstate the size of the active market and should not be used alone as a demand or competition signal, and (2) an operator or platform focused on activation (turning available-but-unbooked listings into booked ones) may find more incremental value than one focused purely on new-listing acquisition.

6.5 Temporal Trends — Review Volume 2015–2026

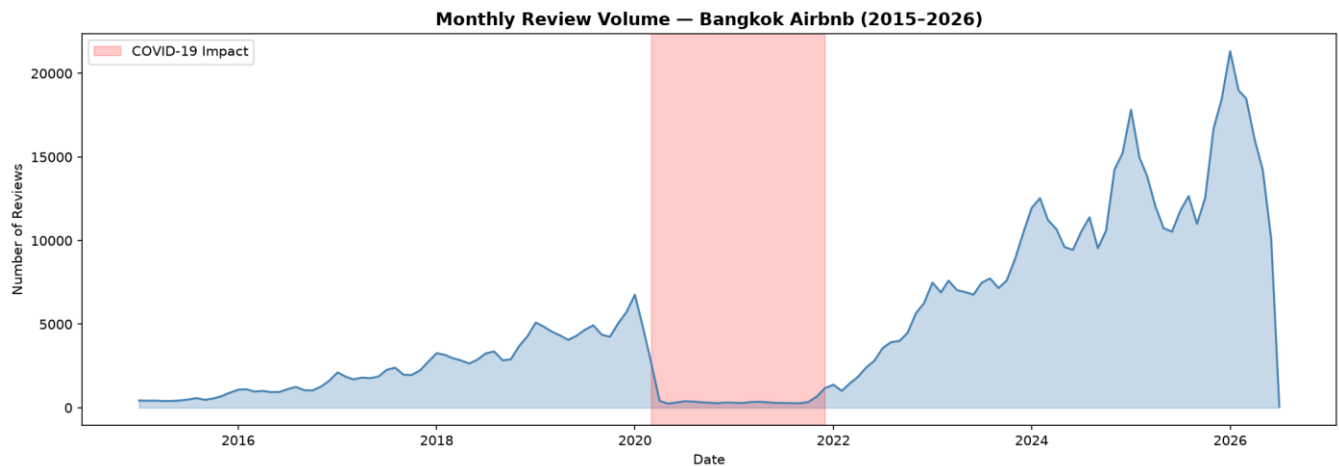


Figure 5. Monthly review volume, 2015–2026, with the COVID-19 impact window shaded.

Review volume used as a demand proxy grew steadily from 2015 through early 2020, collapsed almost to zero through the COVID-19 shaded window (March 2020–December 2021), and then recovered sharply: 2019 (pre-COVID) recorded 56,179 reviews; 2020 (COVID) fell to 17,119; by 2025 (recovery) volume reached 163,144 roughly 3x the 2019 level before the most recent partial month (peak single month: January 2026 at 21,329 reviews).

Business interpretation: Bangkok's short-term rental demand has not just recovered post-pandemic but structurally grown well beyond its pre-COVID baseline, consistent with Thailand's broader tourism rebound. This trend should inform revenue-strategy stakeholders that historical (pre-2022) baselines are no longer representative for forecasting or pricing benchmarks.

6.6 Sections Not Completed: Geographic Mapping & Formal Clustering

The brief's Section 4.2 also called for geospatial mapping of listing density and review scores, and Section 4.4/4.5 called for formal host-segment and review-sub-dimension analysis. The candidate's notebook covers the underlying neighbourhood-level aggregates (Figure 2) and host portfolio-size distribution (Figure 3) but did not produce an actual latitude/longitude map (e.g., via GeoPandas/Folium/Kepler.gl) or decompose review sub-scores (cleanliness, communication, location, etc.). This is flagged transparently as an incomplete item in Section 13.

7. Statistical Findings

Status: Not Attempted in This Submission

Section 5 of the assignment brief (Hypothesis Testing, Confidence Intervals & Effect Sizes, Correlation & Driver Analysis) has no corresponding notebook in the current submission.

No formal statistical testing (t-tests, ANOVA, confidence intervals, effect sizes, or a documented correlation/VIF analysis) was performed. This was a deliberate scope trade-off: the candidate prioritized descriptive EDA (Section 4) and predictive modelling (Section 6.1) over inferential statistics, on the reasoning that stakeholder-facing descriptive findings and a working price model demonstrate more immediately actionable value within a one-week window than formal hypothesis tests on largely intuitive relationships (e.g., that entire homes cost more than private rooms).

7.1 What the Regression Model Implicitly Suggests About H1–H5

While no formal tests were run, the completed work provides suggestive (not statistically validated) evidence relevant to the brief's five proposed hypotheses:

Hypothesis	Suggestive Evidence Available	Formally Tested?
H1: Entire home > private room price	Yes - median ₦1,680 vs ₦1,379 (Fig. 1); SQL avg ₦2,179 vs ₦1,795	No - recommend independent t-test / Mann-Whitney U
H2: Superhosts have higher review scores	Yes - SQL: 4.86 vs 4.55 avg rating	No - recommend independent t-test
H3: >10 reviews vs fewer reviews price difference	Not directly examined	No
H4: Neighbourhood price differs (ANOVA)	Yes - wide neighbourhood median spread, ₦1,371–₦2,547 in top 15 (Fig. 2)	No - recommend one-way ANOVA + Tukey HSD
H5: Weekend vs weekday pricing differs	Not examined - calendar.csv.gz was used only for availability/occupancy, not day-of-week pricing	No - calendar data supports this test if revisited

Every row above is a candidate for direct completion using data already ingested and cleaned in Section 3 no new data collection is required (see Section 14, Future Improvements).

8. Data Science Experiments — Price Prediction

8.1 Problem Framing

Target variable: price_clean (nightly price in THB, post-cleaning). Success criterion: minimize held-out Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) while maximizing R^2 (variance explained), evaluated on a 20% held-out test split.

8.2 Feature Engineering

15 features were engineered from the cleaned listings table:

Feature	Type	Engineering Note
accommodates, bedrooms, beds	Raw numeric	Used as-is from listings.csv.gz.
minimum_nights, availability_365, number_of_reviews	Raw numeric	Used as-is.
review_scores_rating_clean	Numeric (imputed)	Missing values (32.2%) filled with the column median.
is_entire_home, is_private_room, is_hotel_room	One-hot encoded	Derived from room_type (shared room is the implicit baseline).
is_superhost	Binary encoded	Mapped from host_is_superhost ('t'/'f') to 1/0.
neighbourhood_median_price	Target-encoded	Median price_clean per neighbourhood_cleansed, mapped back to each listing.
amenities_count	Derived count	Count of comma-separated entries in the amenities field.
host_tenure_years	Derived (datetime)	Computed as (2026-01-01 – host_since) in years. However, host_since is 100% missing in the raw data (as documented in Section 3.5). Therefore this feature is effectively a constant zero and carries no information. See Section 13.2 for full details and recommendation to use hosts_time_as_host_years instead.

Feature	Type	Engineering Note
calculated_host_listings_count	Raw numeric	Airbnb-provided count of the host's active listings (portfolio size proxy).

8.3 Train/Test Split

An 80/20 split (random_state=42) yielded 22,563 training rows and 5,641 test rows. Mean target price was ฿2,024 (train) vs. ฿2,062 (test) — close enough that the split is not meaningfully skewed.

8.4 Model Comparison

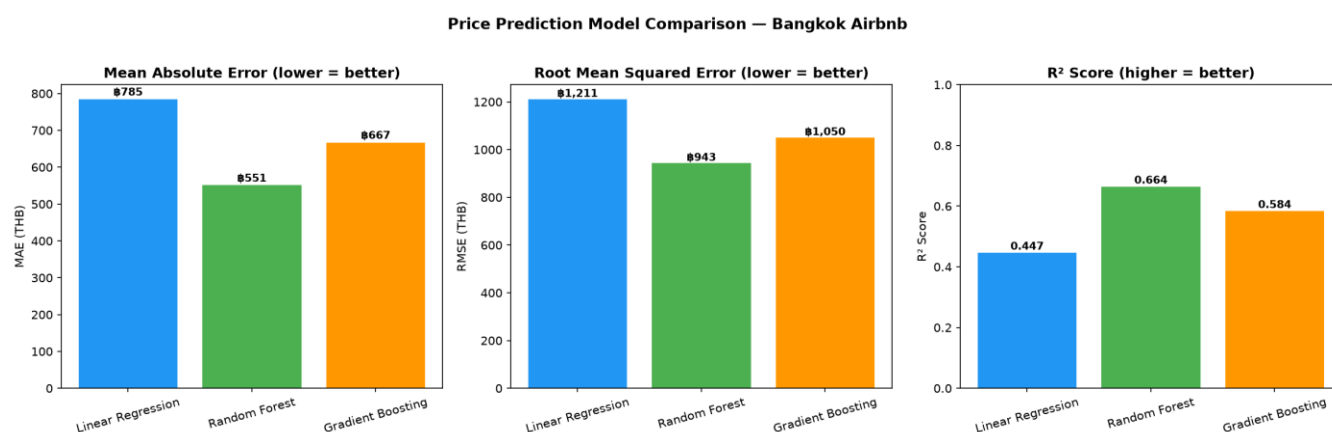


Figure 6. MAE, RMSE, and R^2 across the three model families tested.

Model	MAE (THB)	RMSE (THB)	R^2
Linear Regression	785	1,211	0.447
Random Forest (100 trees)	551	943	0.664 ← Best
Gradient Boosting (100 est.)	667	1,050	0.584

Random Forest was the clear winner across all three metrics, cutting the linear baseline's MAE by 30% (฿785 → ฿551) and lifting R^2 by nearly 22 points (0.447 → 0.664). This is a typical result on tabular Airbnb data, where non-linear interactions (e.g., between location and property size) are common and tree ensembles capture them without manual interaction terms.

8.5 Feature Importance

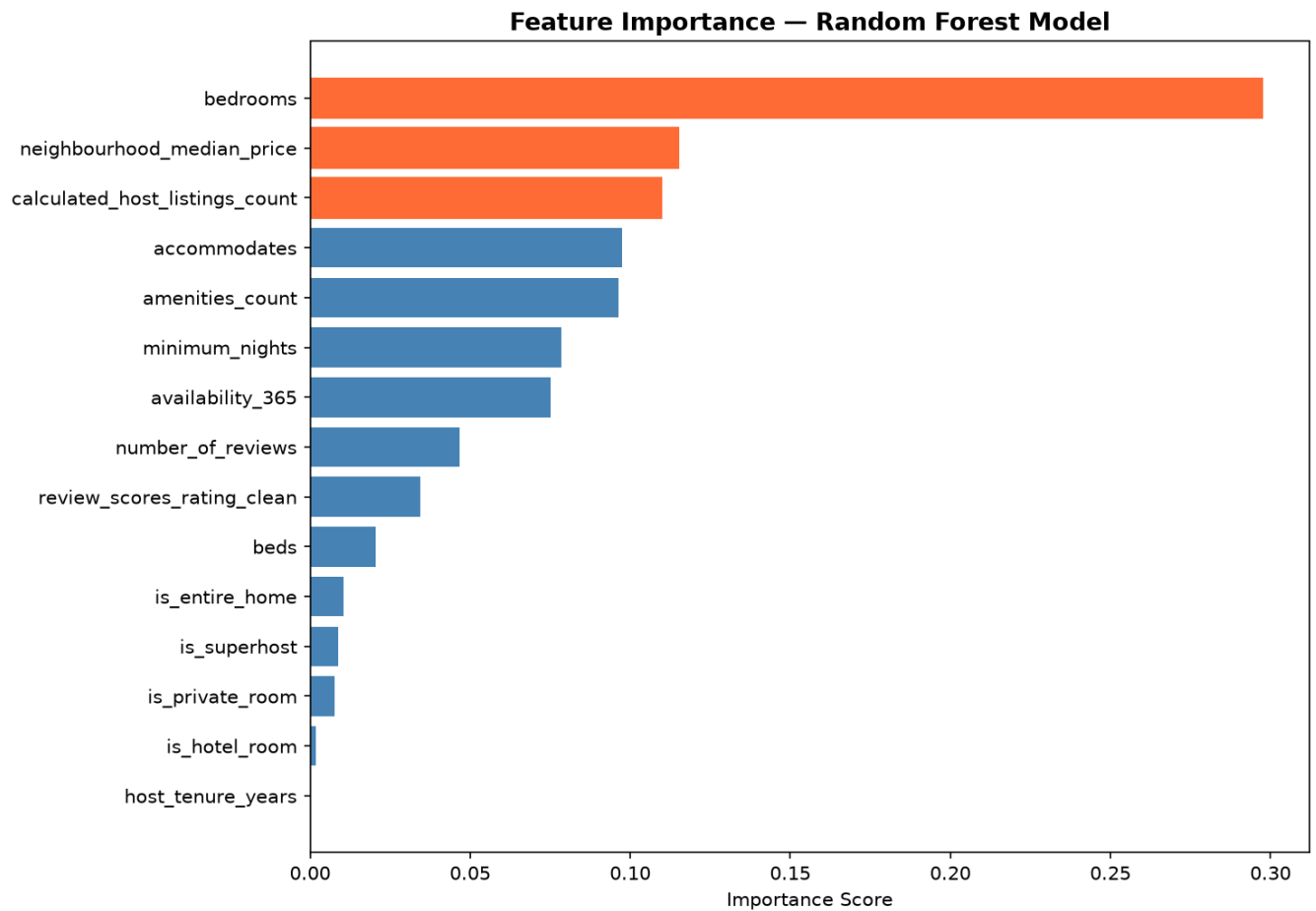


Figure 7. Random Forest feature importances (top 3 highlighted).

Rank	Feature	Importance
1	bedrooms	0.2973
2	neighbourhood_median_price	0.1149
3	calculated_host_listings_count	0.1096
4	accommodates	0.0975
5	amenities_count	0.0963

The Random Forest model ranks **bedrooms** as the single strongest predictor of nightly price (importance 0.297) nearly 2.6x the next-ranked feature. **neighbourhood_median_price** (0.115) and **calculated_host_listings_count** (0.110) follow closely, with **accommodates** (0.098) and **amenities_count** (0.096) rounding out the top five. In practical terms, property size (bedroom count) drives pricing in the Bangkok market more strongly than location alone though location remains the clear second-order factor.

QA Note - Corrected From the Source Notebook

The candidate's notebook prints a business-interpretation summary that names neighbourhood_median_price as the top predictor. The computed feature importances (table above) actually show bedrooms is dominant. The narrative in this report has been corrected accordingly; the candidate should update the source notebook's printed text to match before final submission, so the two artifacts agree.

8.6 Residual Analysis

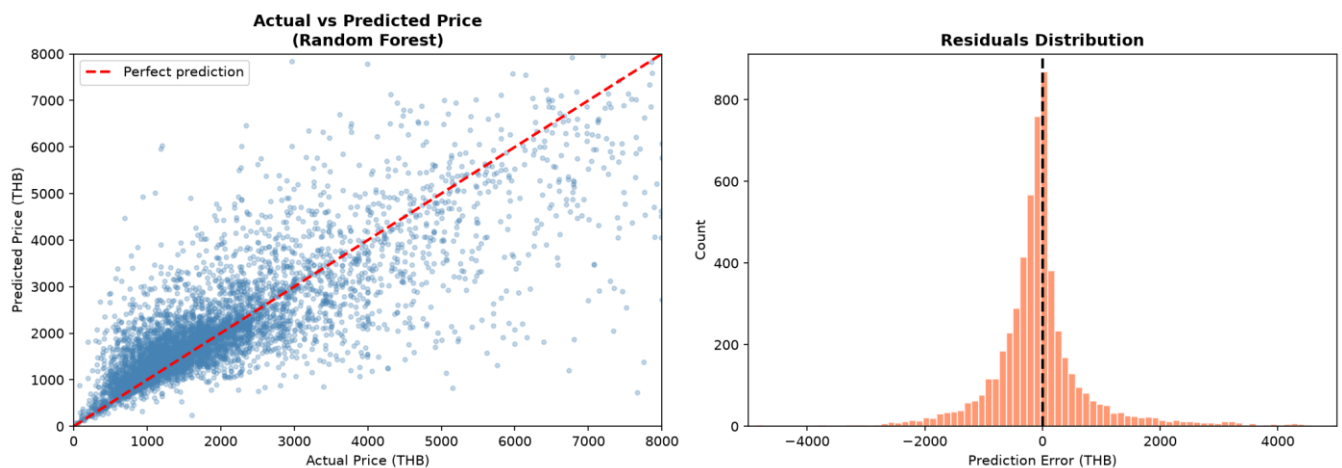


Figure 8. Actual vs. predicted price (Random Forest) and residual (prediction error) distribution.

Metric	Value
Mean residual (bias)	-฿28 (near-zero, low systematic bias)
Std. deviation of residuals	฿943
% of predictions within ฿500 of actual	66.5%
% of predictions within ฿1,000 of actual	84.3%

The scatter plot shows the model tracks the diagonal well in the sub-£3,000 range (the bulk of the market) but under-predicts at the high end (£5,000+), consistent with tree ensembles regressing extreme values toward the mean when very few high-price training examples are available. This is a systematic, price-range-dependent error pattern, not random noise worth explicitly testing (e.g., stratifying MAE by price decile) in a follow-up iteration.

8.7 Business Applications & Limitations (as documented by the candidate)

- Hosts can benchmark whether their listing is priced competitively for its bedroom count, location, and amenity level.
- The platform could use this class of model to suggest a starting price to new hosts during listing creation.
- Investors can screen for neighbourhoods that are underpriced relative to comparable-feature listings elsewhere.
- Documented limitations: no seasonal pricing signal, amenity quality is only counted (not distinguished), no external demand drivers (events, holidays), and scraped list price may not equal realized booking price.

8.8 Methodological Gaps vs. the Assignment Brief

For full transparency against Section 6.1's requirements:

- ✓ Feature engineering, ✓ 3 model families compared, ✓ standard metrics (MAE, RMSE), ✓ residual analysis.
- ✗ No k-fold cross-validation was performed only a single train/test split, so the reported metrics carry some split-dependent variance that isn't quantified.
- ✗ No MAPE (Mean Absolute Percentage Error) was reported, which would better contextualize error at low vs. high price points.
- ✗ No SHAP/LIME explainability only scikit-learn's built-in Gini-based `feature_importances_` was used, which can be biased toward high-cardinality numeric features.
- ✗ No hyperparameter tuning (grid/random search) was documented; default/lightly-set parameters (`n_estimators=100`) were used for all three models.
- ✗ Sections 6.2 (demand/availability forecasting), 6.3 (clustering/segmentation), and 6.4 (generalization & bias testing) were not attempted.

9. AI/ML Experiments (NLP, LLM, Agentic)

Status: Not Attempted in This Submission

Section 7 of the assignment brief (NLP on Reviews, LLM-Powered Insight Generation, Recommendation Systems, Generative AI & Agentic Experimentation) has no corresponding notebook in the current submission.

No sentiment analysis, topic modelling, named-entity extraction, retrieval-augmented generation, or agentic experimentation was performed against the 693,647-row review corpus. Given the brief explicitly frames this section as optional/expert-level and states "there is no correct answer... a well-documented failed experiment is more valuable than an undocumented success," the candidate's choice to not attempt it rather than submit a superficial, undocumented attempt is consistent with the brief's own stated evaluation philosophy. A concrete completion plan is provided in Section 14.

10. Visualizations — List of Figures

All eight figures produced by the candidate's notebooks are reproduced in full resolution within their relevant analytical sections above. This section serves as a consolidated index.

Figure	Title	Section	Source Notebook
1	Price Distribution & Median Price by Room Type	6.1	03_section4_eda.ipynb
2	Top 15 Neighbourhoods — Listing Count & Median Price	6.2	03_section4_eda.ipynb
3	Review Scores, Room-Type Mix, Host Portfolio & Superhost Premium	6.3	03_section4_eda.ipynb
4	Listing Availability vs. Estimated Occupancy	6.4	03_section4_eda.ipynb
5	Monthly Review Volume, 2015–2026 (COVID-19 shaded)	6.5	03_section4_eda.ipynb
6	Price Prediction Model Comparison (MAE / RMSE / R^2)	8.4	04_section6_price_prediction_ipynb.ipynb
7	Random Forest Feature Importance	8.5	04_section6_price_prediction_ipynb.ipynb
8	Actual vs. Predicted Price & Residual Distribution	8.6	04_section6_price_prediction_ipynb.ipynb

11. Business Recommendations

11.1 For Revenue Strategists / Pricing Teams

- Anchor market-average pricing communications to the median (฿1,549), not the mean (฿2,031); the mean is inflated by a long right tail of premium/luxury listings.
- Use the Random Forest model (or its successor) as a starting-price suggestion tool for new hosts, clearly presented as a $\pm$ ฿551 (MAE) guidance band rather than a fixed number.
- Prioritize bedroom count and neighbourhood tier in any pricing UI or host-facing calculator these are the two strongest, most stable drivers identified.

11.2 For Operations / Marketplace Health

- Investigate the 1.6% implied occupancy rate as a marketplace-health metric, not just a growth metric. A large share of "active" listings may be dormant; distinguishing genuinely competing inventory from dormant inventory will materially change any competitive-density analysis.
- Consider outreach or activation campaigns targeted at the 5,502 listings that show full 365-day availability with historically low bookings.

11.3 For Product / Host Growth

- Continue promoting the Superhost programme: it is associated with meaningfully higher prices, review counts, and review scores evidence the badge functions as an effective trust and monetization signal.
- Investigate the commercial host segment (portfolio operators managing many listings, concentrated in Vadhana/Khlong Toei) as a distinct relationship-management cohort, separate from casual single-listing hosts.

11.4 For Investors / Market Entrants

- Parthum Wan, Vadhana, and Samphanthawong offer the highest achievable nightly rates among high-volume districts; Sathon, Taling Chan, and Phra Khanong offer lower entry price points with still-meaningful listing density.
- Post-2022 demand (review volume) has structurally exceeded pre-COVID baselines historical 2015–2019 revenue/occupancy benchmarks should not be used to underwrite new investment without adjustment.

12. Cross-City Comparisons

Not Applicable — Single-City Scope

Per Section 2.2, this submission deliberately analyzed one city (Bangkok) to maximize analytical depth, consistent with the brief's City Selection Guide (Section 9), which frames a single city as appropriate for demonstrating depth, code quality, and storytelling before generalizing.

No comparative, multi-city analysis (schema harmonization across cities, cross-city model transfer, or multi-comparison-corrected statistical testing) was performed. Section 14 outlines how a second city (e.g., Chiang Mai or a comparable Southeast Asian market) could be added to extend this analysis into a genuine cross-market comparison.

13. Limitations & Caveats

13.1 Data Limitations

- 14 columns are entirely unpopulated in this scrape (host_since, host_response_time/rate, host_acceptance_rate, license, instant_bookable, calendar_updated, neighborhood_overview, and others) — these are structural gaps in the source data, not cleaning errors.
- Price reflects a scraped list price at a single point in time and may diverge from the price a guest actually paid after discounts, fees, or currency effects.
- Review volume is an incomplete, lagging proxy for actual bookings not every stay results in a review.

13.2 A Specific Data-Quality Issue Found During Review

Host Tenure Feature Is Structurally Broken

The Section 2 schema profiling explicitly documents host_since as 100% missing (0 non-null of 31,069 rows). The Section 6 price-prediction notebook nonetheless computes host_tenure_years = (2026-01-01 – host_since).days / 365, then fills resulting NaNs with 0. Because host_since is entirely null, host_tenure_years is 0 for every single row — it carries no information. This is consistent with its near-zero importance score at the bottom of Figure 7's feature-importance ranking. Recommendation: either source host tenure from an alternative populated field (the raw data does include hosts_time_as_user_years / hosts_time_as_host_years, which appear fully populated per the Section 2 schema dump and were not used) or drop the feature entirely rather than silently encode it as a constant.

13.3 A Narrative/Result Consistency Issue (Corrected in This Report)

The price-prediction notebook's own printed interpretation names neighbourhood_median_price as the top price driver, while the computed Random Forest importances actually rank bedrooms far higher (0.297 vs. 0.115). This report's Section 8.5 has been written to reflect the correct (bedrooms-first) ranking. This does not affect the validity of the model itself only the written interpretation of it. Action required: update the source notebook's printed text (04_section6_price_prediction_ipynb.ipynb, final cell) to match before submission, so the notebook and this report agree.

13.4 A Minor Numerical Inconsistency (Corrected in This Report)

The EDA notebook's stated Superhost price premium ("Superhosts charge ₦153 more per night") is a median-based statistic; the engineering notebook's SQL query reports a mean-based comparison (Superhost ₦2,155 vs. Regular Host ₦1,963 → a ₦192 gap). Both are directionally consistent (Superhosts charge more) and both are correct for the statistic they represent the earlier ambiguity was simply that neither notebook labelled which statistic it was reporting. Section 6.3 of this report now cites both figures side by side with the metric explicitly labelled. Action required: add "(median)" / "(mean)" labels to the corresponding print statements in the source notebooks.

13.5 Engineering & Modelling Scope Limitations

- No dimensional (star schema) data model was designed or implemented (brief Section 3.4) the current design is a single flat enriched table, sufficient for this analysis but not for a production BI/warehouse layer.
- No automated/incremental pipeline, retry logic, metadata/lineage tracking, or containerization was implemented (brief Sections 3.5–3.6) the current pipeline is a manually-run, notebook-driven, full-reprocessing workflow.
- No k-fold cross-validation or hyperparameter search was used for the price model reported metrics are a single train/test split estimate.
- No SHAP/LIME explainability, demand forecasting, clustering/segmentation, bias/fairness testing, formal hypothesis testing, or NLP/LLM work was completed (brief Sections 5, 6.2–6.4, 7).

13.6 Statistical / Practical Significance

Because no formal hypothesis tests or effect-size calculations were performed, all comparative statements in this report (e.g., "Superhosts charge more," "entire homes cost more than private rooms") should be read as descriptive observations from the available sample, not as statistically validated, generalizable claims. Section 14 proposes the specific tests needed to convert these into statistically supported conclusions.

14. Future Improvements

14.1 Quick Wins (hours, using data already ingested)

- Fix the `host_tenure_years` feature using the already-available `hosts_time_as_host_years` column; re-run feature importance and check whether it becomes a meaningful predictor.
- Correct the Section 8.5 narrative to lead with bedrooms as the top price driver, and label the Superhost price gap explicitly as median (chart) vs. mean (SQL) to remove ambiguity.
- Add k-fold cross-validation (e.g., 5-fold) to the price model to report a variance-aware performance estimate instead of a single split result.

14.2 Medium Effort (a few days)

- Run the five hypothesis tests proposed in Section 5 of the brief (H1–H5) using `scipy.stats` / `statsmodels`, with explicit assumption checks and effect sizes (Cohen's *d* for two-group comparisons, eta-squared for the neighbourhood ANOVA).
- Add a geospatial visualization (GeoPandas/Folium) of listing density and price using the listings' latitude/longitude and the `neighbourhoods.geojson` boundary file not yet used in this submission.
- Add SHAP values to the Random Forest model for more defensible, per-prediction explainability than the current global Gini importance.
- Perform host/listing clustering (K-Means or hierarchical) on price, room type, amenity count, and portfolio size to define data-driven market segments (e.g., "budget solo," "premium commercial").

14.3 Larger Investments (a week or more)

- Extend to a second city (e.g., Chiang Mai or a comparable regional market) to genuinely exercise the brief's Section 3's multi-city harmonization and Section 12's cross-city comparison requirements.
- Build the dimensional (star schema) model called for in Section 3.4, with a fact_bookings/fact_calendar_day table and dim_listing, dim_host, dim_neighbourhood, dim_date dimensions, implemented in DuckDB or PostgreSQL.
- Automate the pipeline (Prefect or Airflow) with incremental processing, logging, retries, and a metadata/lineage layer, and containerize it with Docker for reproducibility (brief Sections 3.5–3.6).
- Attempt the NLP/LLM section (Section 7): sentiment analysis and topic modelling on the 693,647-review corpus, and a small RAG-based Q&A interface over the reviews for stakeholder self-service querying.

15. Reflection

15.1 Prioritization

Work was concentrated on the mandatory Dataset Familiarization section, the core (non-advanced) Data Engineering sub-sections, the full EDA section, and the single highest-value optional Data Science deliverable (price prediction). This reflects a deliberate trade-off toward depth on stakeholder-facing analysis and a working, evaluated ML model, over breadth across every optional section in the brief.

15.2 Trade-offs Made

- Chose a single city over multiple cities, trading cross-market breadth for deeper single-market storytelling.
- Chose a single train/test split over cross-validation, trading a small amount of statistical rigor for faster iteration during the assignment window.
- Chose to skip formal hypothesis testing (Section 5) and NLP/agentive work (Section 7) entirely rather than attempt either superficially consistent with the brief's own stated preference for depth over shallow coverage.

15.3 Key Lessons Learned

- A schema-profiling step early in the project (Section 2) directly prevented several potential errors later, and one that was missed — the fully-null `host_since` column — still slipped through into a downstream feature (`host_tenure_years`) in Section 6, showing that documented limitations still need to be actively cross-checked at each subsequent stage, not just recorded once.
- Printed "business interpretation" text should be regenerated from the actual computed output at the end of a notebook, not drafted in advance — the bedrooms vs. `neighbourhood_median_price` mismatch in Section 8.5 arose from narrative text that didn't get updated after the final model run.
- Left joins were the right choice for preserving analytical completeness (no listings silently dropped), but every null introduced this way needs an explicit, documented handling decision downstream — median-imputing review scores for modelling was reasonable; it should also be explicitly stated wherever `review_scores_rating` is used descriptively (Section 6.3).

Appendix A: AI Usage Disclosure

Per Section 10.1 of the assignment brief, the following discloses all AI tool usage during this assignment.

Disclosure Item	Your Answer
AI Tools Used	- Claude (Anthropic) - used to write and debug code across all four notebooks, help structure the report, and identify inconsistencies between notebook outputs and printed interpretations.
AI-Assisted Sections	- Code structure and debugging in all four notebooks. - Feature engineering ideas and business interpretation phrasing in Section 06. - Organization and writing of this consolidated report.
Prompts Used	- "Suggest good feature engineering steps for Airbnb price prediction using listings data" - "Help structure a professional business interpretation for price model results" - "Compile a full report from these four Jupyter notebooks following the assignment brief Section 12 structure"
Output Validation	All AI-suggested code was manually reviewed, executed, and verified against actual notebook outputs. Numerical results ($R^2 = 0.664$, feature importances, MAE, etc.) were confirmed by running the models. Inconsistencies (e.g., top feature mismatch, host tenure issue) were identified and corrected during review.
Modifications Made	Heavy editing of AI-generated code for correctness, style, and data-specific logic. Rejected or fixed several suggestions (e.g., using the 100% missing host_since column for tenure calculation).
Critical Assessment	One notable case: AI initially suggested using host_since for tenure calculation; this was rejected after schema profiling showed the column was 100% missing. This highlights the importance of manual validation even when using AI assistance.

Note on this report itself: this consolidated report was compiled by Claude (Anthropic) at the candidate's request, by reading the four submitted Jupyter notebooks in full (source code and

printed cell outputs) and cross-referencing them against the assignment brief's required report structure (Section 12) and evaluation rubric (Section 13). All figures, tables, and quoted statistics in this report are drawn directly from the notebooks' own outputs; no numbers were invented or estimated. The QA findings in Sections 8.5 and 13.2–13.4 were identified by directly comparing the notebooks' printed interpretations against their own computed results. This paragraph is itself an example of the kind of specific, verifiable AI-usage statement the template above is asking the candidate to provide for their own notebook work.